BCA ? ARTIFICIAL INTELLIGENCE ? SECTION A

HOUSE PRICE ESTIMATION

A machine-learning approach to property value prediction

LINEAR REGRESSION

SYNTHETIC DATA

Machine Learning Project ? ML-004

PROJECT OVERVIEW

Features ? Estimated price

PROGRAM

BCA (Artificial Intelligence), Section A

MENTOR

Harsh Shrivastav

PROJECT TEAM

The people behind the project

Team roster and student identifiers

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Property prices depend on multiple factors

A consistent estimate requires learning how property features relate to price.

01 Price variation

Area, location, bedrooms, bathrooms and age can all influence value.

02 Manual inconsistency

Human estimates may vary from one property or evaluator to another.

03 Project objective

Use machine learning to estimate price from selected property features.

GOAL **Build and evaluate a Linear Regression model; identify useful price-related patterns.**

A controlled experiment with synthetic records

Dataset and split details from the project

500 property records

80:20 train / test split

42 random state

INPUT FEATURES

Area_sqft ? Bedrooms ? Bathrooms ? Age_years ? Location_Category

PREDICTION TARGET

Price_INR

400 training records ? 100 test records

Python tools support the end-to-end workflow

Core libraries used for data preparation, modelling and evaluation

PY

Python

Project implementation

Pd

Pandas

Data handling

Np

NumPy

Numerical operations

Mp

Matplotlib

Visualisation

Sk

Scikit-learn

Model + evaluation

DATA ? PREPARE ? TRAIN ? EVALUATE

Preprocessing feeds a single model pipeline

Numerical and categorical features are handled before prediction.

01 Inspect & select

Prepare data and choose relevant features

?

02 Transform inputs

Encode location with OneHotEncoder

?

03 Split data

80% train ? 20% test

?

04 Fit model

Linear Regression learns feature relationships

Scikit-learn Pipeline ColumnTransformer + OneHotEncoder(handle_unknown="ignore") + Linear Regression

05 ? RESULTS

Evaluation on the synthetic test set

Reported metrics describe model performance on this experiment only.

MAE

?382,443

Average absolute prediction error

RMSE

?490,665

Penalises larger errors more strongly

R? SCORE

0.985

Reported goodness of fit

IMPORTANT **Synthetic-data results do not represent real-world market accuracy.**

Visual checks help inspect model behaviour

Use plots to assess relationships, predictions and residuals.

01 **Area vs Listed Price**

Inspect the relationship between area and price.

02 **Actual vs Predicted**

Check whether predictions track observed test values.

03 **Error Distribution**

Review how prediction errors are spread.

04 **Model Coefficients**

Compare direction and relative feature influence.

INTERPRETATION RULE Use actual plots to confirm patterns; no visual result is inferred here.

Promising baseline; real-world validation remains

Key limitations and practical next steps

Current limitations

Synthetic records; broad location categories; property condition and amenities are not included.

Future scope

Use verified property data, richer features, cross-validation and model comparisons.

DEPLOYMENT IDEA

A Streamlit or Flask interface could present estimates with clear uncertainty.

THANK YOU!

Questions and feedback welcome.

KEY TAKEAWAY

Linear Regression offers a clear starting point for price estimation.

Validate with real data before practical use.

BCA (Artificial Intelligence), Section A ? Mentor: Harsh Shrivastav

References: Scikit-learn Documentation ? Pandas Documentation ? Matplotlib Documentation